



Investment Society
Imperial College London

Lecture 7

Commodities & Real Estate

Securities Education Certificate

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Slide deck inspired by Dr. Euan Doidge and Prof. Anthony Barrett (Department of Chemistry, Imperial College London)
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Background

Some context to introduce you to a concept

Fact

You will need to recall this information in the exam

Model Example

A fully detailed breakdown of an example (question)

Try it yourself

It's your turn to give this problem a go and test your knowledge!

Equation sheet

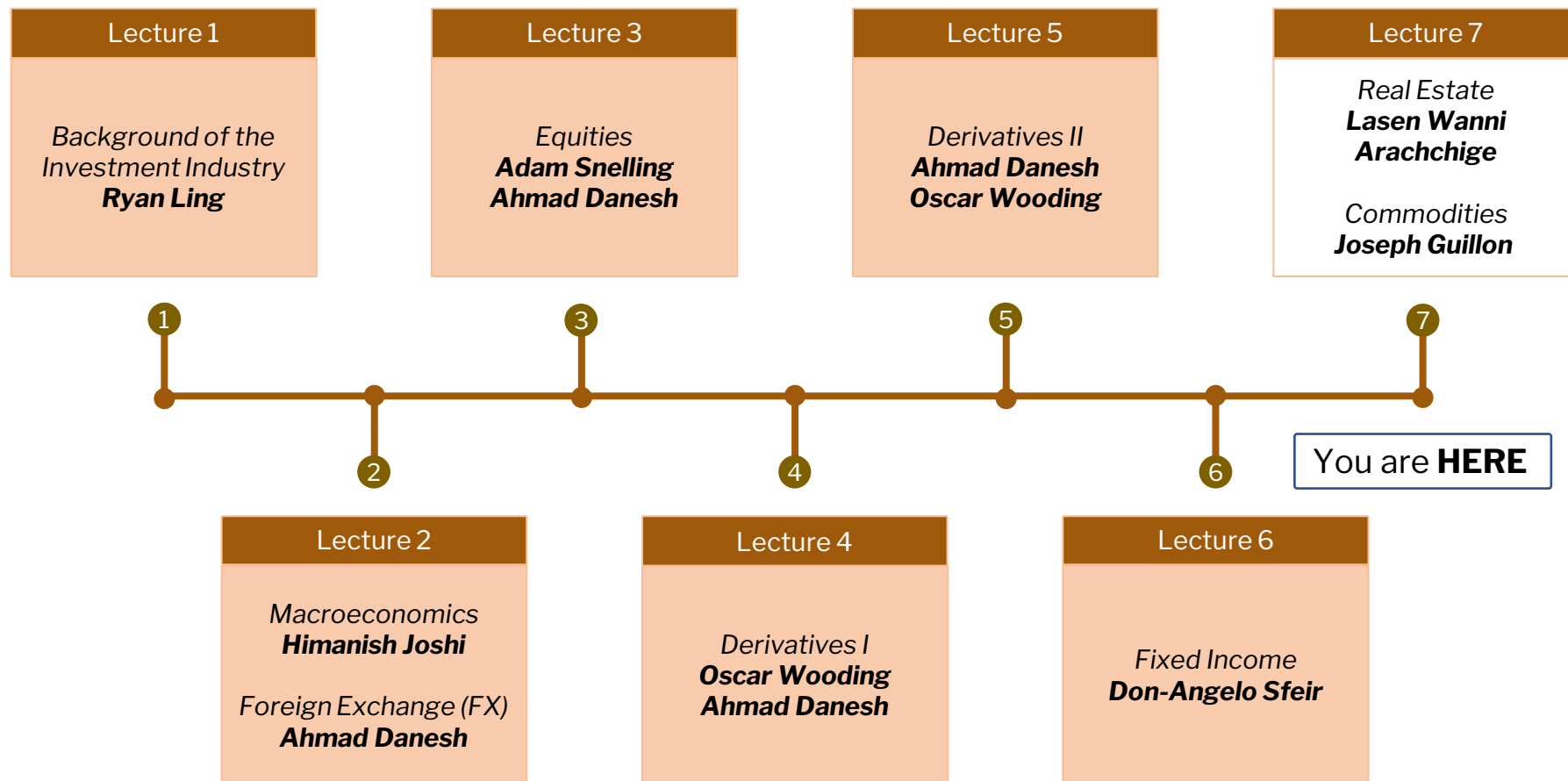
Any equations that you may need for the exam are provided

Taking it further

(non-examinable) stuff to show you where this concept can be applied

The slides are colour-coded to aid your understanding of the course and ensure that you are familiar with the demands of the content provided in each slide.

Lecture Timeline



Let's begin...



Lasen Wanni Arachchige

Lecturer

*Senior Analyst, QT Capital
Real Estate*

Lecture 7

Real Estate

Introduction to Real Estate

What do we actually mean by **Real Estate**?

☞ Simple: a property consisting of land **OR** buildings → **Real Estate**

☞ Complex: Land and any permanent improvements to the land (natural/man made) → **Real Estate**

Features
Real Estate

Water

Trees

Minerals

Buildings

Fences



Your notes

Types of Real Estate

What types of Real Estate exist?



Residential Real Estate

Real property (land and any buildings on it) used for residential purposes.

Family residences

Detached houses

Townhouses

Apartments



Commercial Real Estate

Any property used exclusively for business purposes.

Office Buildings

Hotels

Shopping Centres

Parking Facilities



Industrial Real Estate

Property used for manufacturing, production, distribution, storage and R&D.

Factories

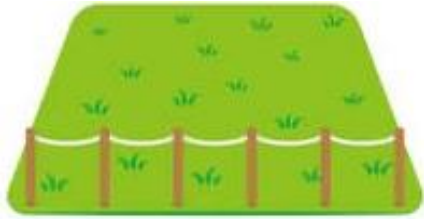
Powerplants

Warehouses

Data Centres

Types of Real Estate

There are more types of Real Estate as well...



Vacant Land

Undeveloped land that has no buildings on it and is not being used.

Raw Land

Natural Fields

Forests



Special Purpose Real Estate

Property used by the public

**Government
buildings**

**Public
libraries**

Cemeteries

**Places of
worship**

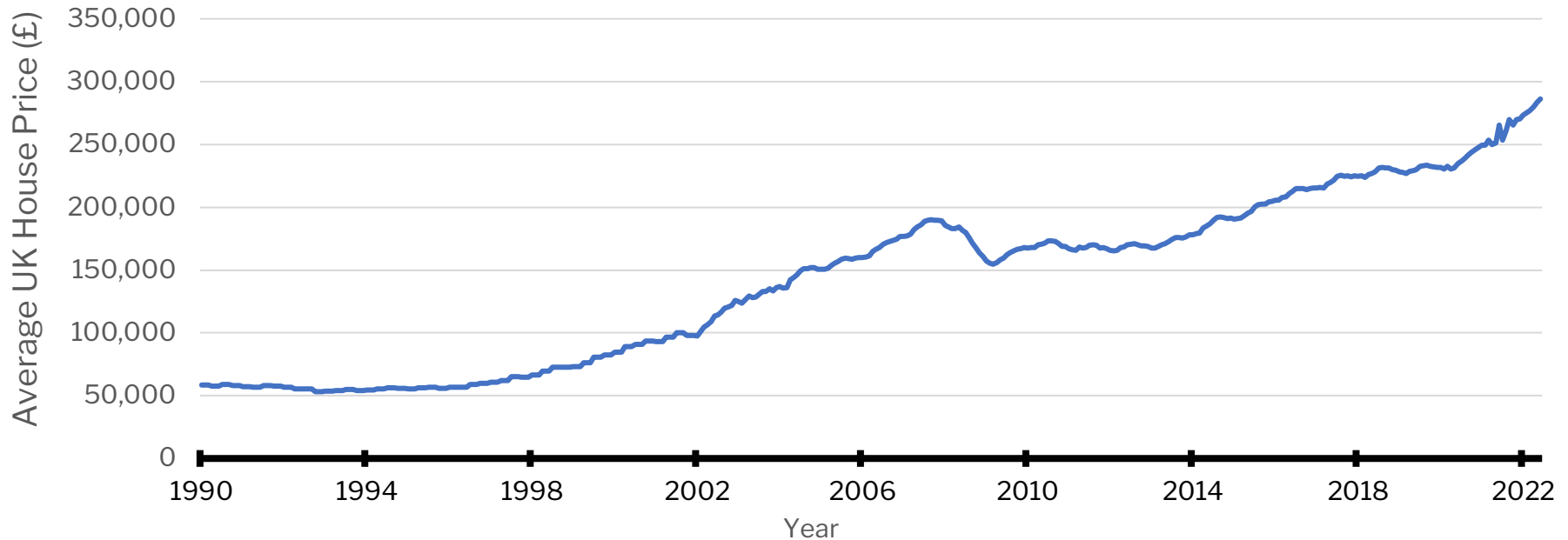
👉 Each type of real estate requires unique analysis for potential investment.

Your notes

Historical Prices

How have Real Estate prices been changing over time?

Average UK House Price, 1990–

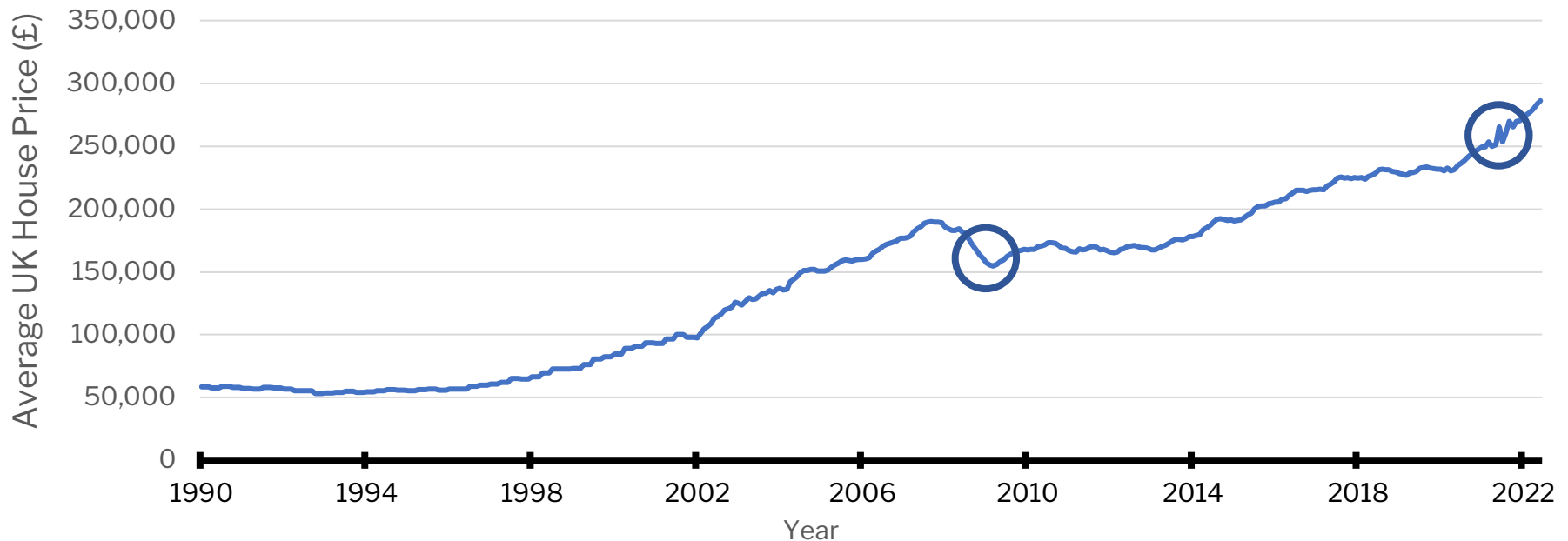


👉 Real estate has long been considered a *safe* investment.

Historical Prices

How have Real Estate prices been changing over time?

Average UK House Price, 1990–



With a few exceptions, house prices tend to demonstrate continuous growth. Even after small hits, the real estate market recovers quickly.

Investing in Real Estate

So, how do we consider whether to invest in Real Estate?



Buy



Don't Buy

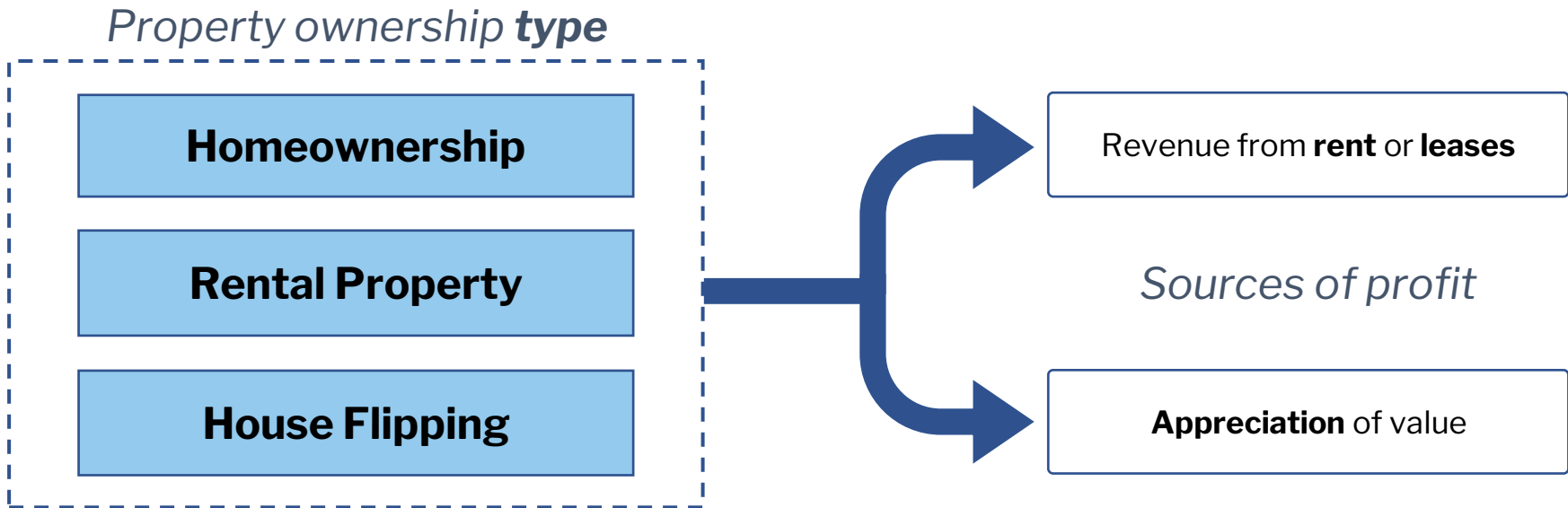
☞ Several factors must be considered when investing in Real Estate (RE).

Your notes

Purchase of Physical Properties

What does the purchase of physical properties entail?

👉 Two main sources of profit when looking at physical properties.



Your notes

Purchase of Physical Properties

Why might someone purchase Real Estate?

Homeownership



Purchase of residential property for the owner to live in.

Most common form of real estate investment

BENEFIT

Appreciation in house value

Rental Property



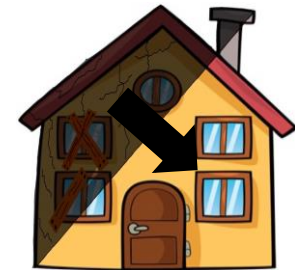
Purchase of property with the intent of leasing it out to tenants.

Owner becomes landlord with an income through rent.

BENEFIT

Appreciation in house value

House Flipping



Purchase of property with the intention of holding it for a short period (3 – 4 months) of time and quickly selling.

Repair and update
OR Hold and resell

Purchase of Physical Properties

What are the advantages of each purchase type?

Homeownership

ADVANTAGE

Capital Appreciation (Increase)

ADVANTAGE

Diversifies the portfolio

ADVANTAGE

Can be bought using leverage
(e.g. mortgage)

Rental Property

ADVANTAGE

Capital Appreciation (Increase)

ADVANTAGE

Provides regular income

ADVANTAGE

Maximise capital
(through leverage)

House Flipping

ADVANTAGE

Ties up capital for a short time

ADVANTAGE

Can offer quick returns

👉 These factors vary based on e.g. geography or market momentum.

Your notes

Purchase of Physical Properties

What are the disadvantages of each purchase type?

Homeownership

DISADVANTAGE

Low Risk **BUT** Low Reward

DISADVANTAGE

Requires Large Capital → Illiquid

DISADVANTAGE

Local factors affect prices

Rental Property

DISADVANTAGE

No income from vacancies

DISADVANTAGE

Tedious to manage tenants

DISADVANTAGE

Damage to property by tenants

House Flipping

DISADVANTAGE

Needs good market knowledge

DISADVANTAGE

Requires *accurate* valuations

DISADVANTAGE

Market can cool *unexpectedly*

👉 **AGAIN** These factors vary based on e.g. geography or market momentum.

Your notes

Calculating ROI

What do we mean by an ROI?

👉 **ROI** stands for **R**eturn **O**n **I**vestment.

👉 ROI → the money earned on an investment *after* all expenses are deducted.

👉 You can use two methods to calculate ROI:

Cost method

Out-of-pocket method

👉 We'll be covering them separately in the following slides.

Your notes

Calculating ROI

How is the ROI calculated using the cost method?

👉 ROI → the money earned on an investment *after* all expenses are deducted.

Cost method

👉 Divide the investment gain in a property by the property's initial cost.

$$\text{Cost Method ROI (\%)} = \frac{\text{Net Investment Gain}}{\text{Initial Cost}} \times 100 = \frac{\text{Investment Gain} - \text{Investment Cost}}{\text{Total Costs}} \times 100$$

👉 Assumes property is purchased outright with cash **AND** without financing

Your notes

Calculating ROI

Let's try an example to see how the cost method of ROI works...

Cost method for home ownership / house flipping

$$\text{Cost Method ROI (\%)} = \frac{\text{Net Investment Gain}}{\text{Initial Cost}} \times 100 = \frac{\text{Investment Gain} - \text{Investment Cost}}{\text{Total Costs}} \times 100$$

Cost
Method



Initial Value: £100,000

Repairs + improvements: £50,000

Final Value:
£200,000

$$\begin{aligned} \text{Investment Gain} - \text{Investment Cost} &= £200,000 - (£100,000 + £50,000) \\ &= \mathbf{£50,000} \end{aligned}$$

$$\begin{aligned} \text{Total Cost} &= £100,000 + £50,000 \\ &= \mathbf{£150,000} \end{aligned}$$

$$\text{Cost Method ROI (\%)} = \frac{£50,000}{£150,000} \times 100 = 33\%$$

Calculating ROI

What is ROI and how is it calculated using the cost method ?

Cost method for a Rental Property

$$\text{Cost Method ROI (\%)} = \frac{\text{Net Investment Gain}}{\text{Initial Cost}} \times 100 = \frac{\text{Investment Gain} - \text{Investment Cost}}{\text{Total Costs}} \times 100$$

Cost
Method



Initial Value: £100,000

Rent: £1200 per month

Repairs + improvements: £10,000

Expenses (bills/ maintenance): £250 per month

$$\begin{aligned} \text{Annual Investment Gain} - \text{Annual Investment Cost} &= (12 \times £1,200) - (12 \times £250) \\ &= \mathbf{£11,400} \end{aligned}$$

$$\begin{aligned} \text{Total Cost} &= £100,000 + £10,000 \\ &= \mathbf{£110,000} \end{aligned}$$

$$\text{Annual ROI (\%)} = \frac{£11,400}{£110,000} \times 100 = 10.4\%$$

Calculating ROI

How is the ROI calculated using the out-of-pocket method?

👉 ROI → the money earned on an investment *after* all expenses are deducted.

Out-of-Pocket Method

👉 This method uses **leverage** to purchase property instead of purchasing the property outright.

$$\text{Out-of-Pocket ROI (\%)} = \frac{\text{Current Equity Position}}{\text{Final Value of Property}} \times 100$$

👉 Out-of-pocket method is preferred by most investors as it results in higher ROI.

Your notes

Calculating ROI

What is ROI and how is it calculated using the out-of-pocket method?

Out-of-Pocket method for home ownership / house flipping

$$\text{Out-of-Pocket ROI (\%)} = \frac{\text{Current Equity Position}}{\text{Final Value of Property}} \times 100$$

Out-of-Pocket
Method



Initial Value: £100,000

Repairs + improvements: £50,000

Final Value:
£200,000

Original out-of-pocket expense = £20,000 + £50,000
= £70,000

Current equity position = £200,000 - £70,000
= £130,000

Mortgage used to
finance house with
initial down payment
of £20,000

$$\text{ROI (\%)} = \frac{\text{£130,000}}{\text{£200,000}} \times 100 = 65\%$$

Calculating ROI

What is ROI and how is it calculated using the out-of-pocket method?

Out-of-Pocket method for Rental Property

$$\text{Out-of-Pocket ROI (\%)} = \frac{\text{Current Equity Position}}{\text{Final Value of Property}} \times 100 \quad \frac{\text{Annual Rental Income} - \text{Annual Expenses}}{\text{Original out-of-pocket expenses}} \times 100$$

Out-of-Pocket
Method



Initial Value: £100,000

Rent: £1200 per month

Repairs + improvements: £10,000

Expenses (bills/ maintenance): £250 per month

$$\text{Annual rental income} - \text{Annual expenses} = (12 \times £1,200) - (12 \times £250) - (7,000) \\ = \mathbf{£4,400}$$

$$\text{Original out-of-pocket expenses} = £20,000 + £10,000 \\ = \mathbf{£30,000}$$

$$\text{Annual ROI (\%)} = \frac{£4,400}{£30,000} \times 100 = 14.7\%$$

Mortgage used to finance house with initial down payment of **£20,000**

Annual principal + interest **£7,000**

Calculating ROI

Let's compare both methods of ROI briefly side-by-side.

ROI – Return On Investment – A measure of how much money earned on an investment as a percentage of total cost

Cost method **'Net Capitalisation Rate' (Cap Rate)**

Divide investment gain by property cost

ASSUMPTION

Property is purchased outright with cash i.e. **NO** financing

$$\text{ROI (\%)} = \frac{\text{Net Operating Income (NOI)}}{\text{Initial Cost}} \times 100$$

$$\text{NOI} = \text{Gain on Investment} - \text{Cost of Investment}$$

Out-of-pocket method

Factors in property financing with leverage

Leveraged purchases lead to higher ROI
Preferred by most investors

$$\text{ROI (\%)} = \frac{\text{Net Operating Income (NOI)}}{\text{Original out-of-pocket expenses}} \times 100$$

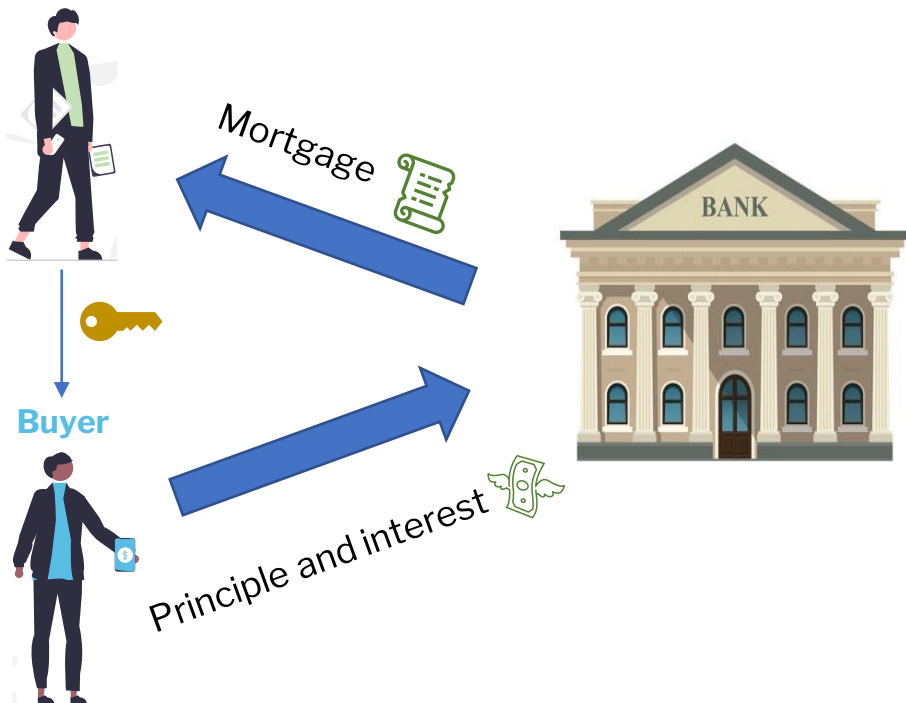
Both methods use similar equations
BUT slight variations to NOI and total costs

Mortgage-Backed Securities

Principle, Interest and Bonds covered in Lecture 6 on fixed income

What is a mortgage-backed security?

Seller



Before Mortgage-Backed Securities

Buyer applies for mortgage which includes interest

Seller receives payment for house
and **buyer** receives keys

Mortgage is kept on bank's books for duration

Bank receives principal and interest from **buyer**

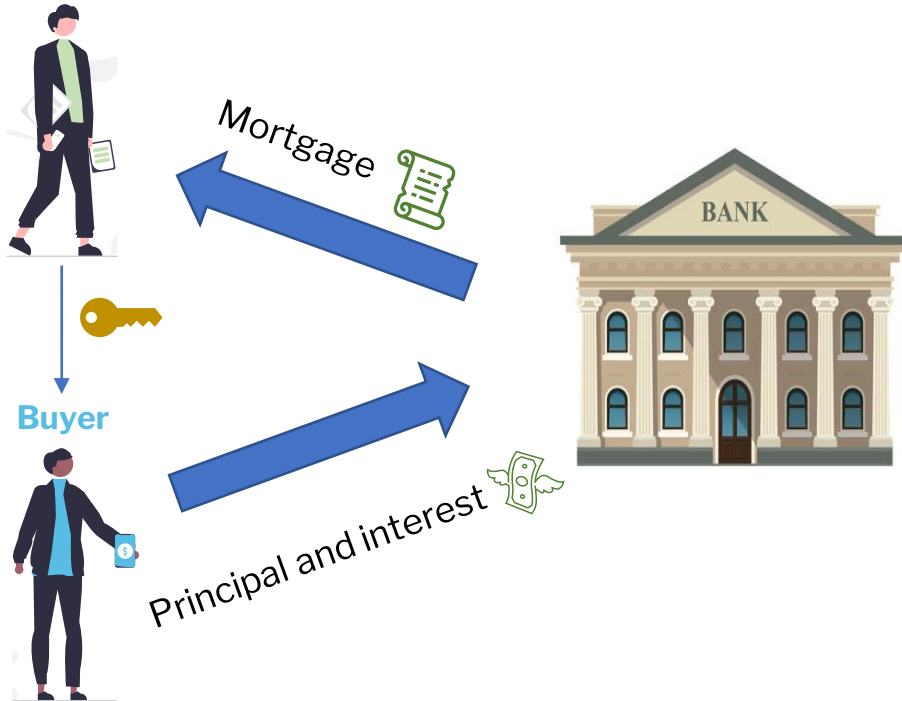
Your notes

Mortgage-Backed Securities

Principle, Interest and Bonds covered in Lecture 6 on fixed income

What is a mortgage-backed security?

Seller



Your notes

Before Mortgage-Backed Securities

PROBLEM

Ties up bank's resources and capital

Opportunity cost associated with lack of funds available to bank

👉 **Opportunity Cost:**
The potential **benefits lost** out when choosing an **alternative** over another.

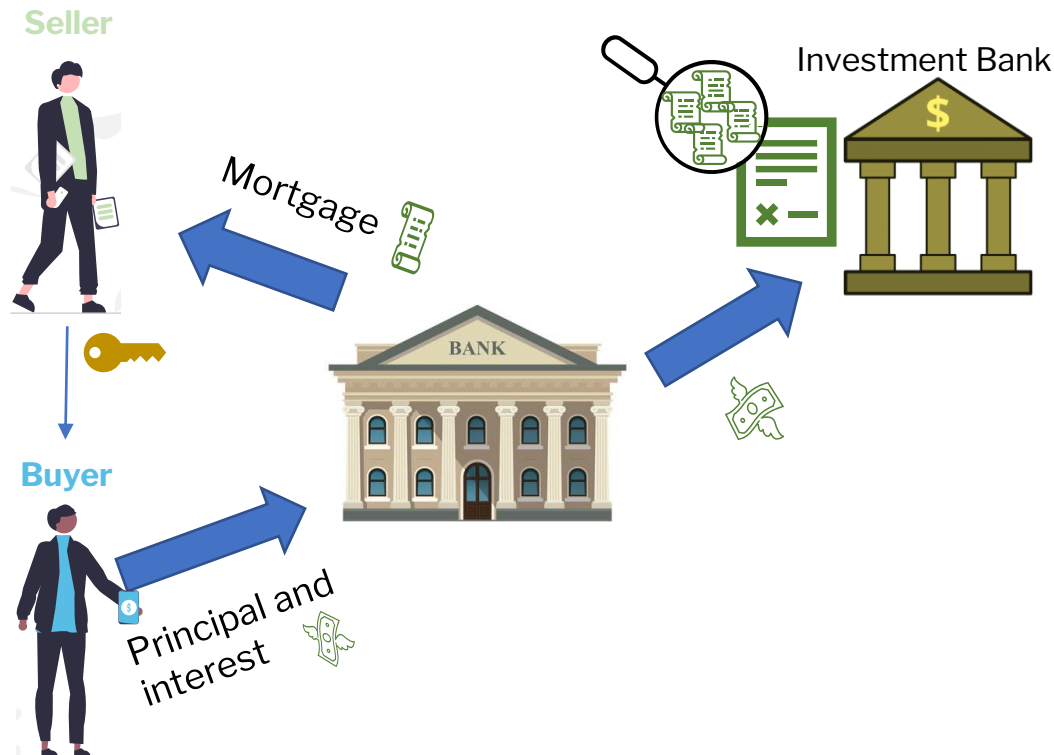
Mortgage-Backed Securities

Principle, Interest and Bonds covered in Lecture 6 on fixed income

What is a mortgage-backed security?

MBS – Mortgage-Backed Security

A type of bond representing an investment in a pool of real estate loans



Mortgage-Backed Securities

Bank bundles lots of mortgages together

This bundle is sold to investment banks in the form of a bond

Investors receive principal and interest paid to bank by seller

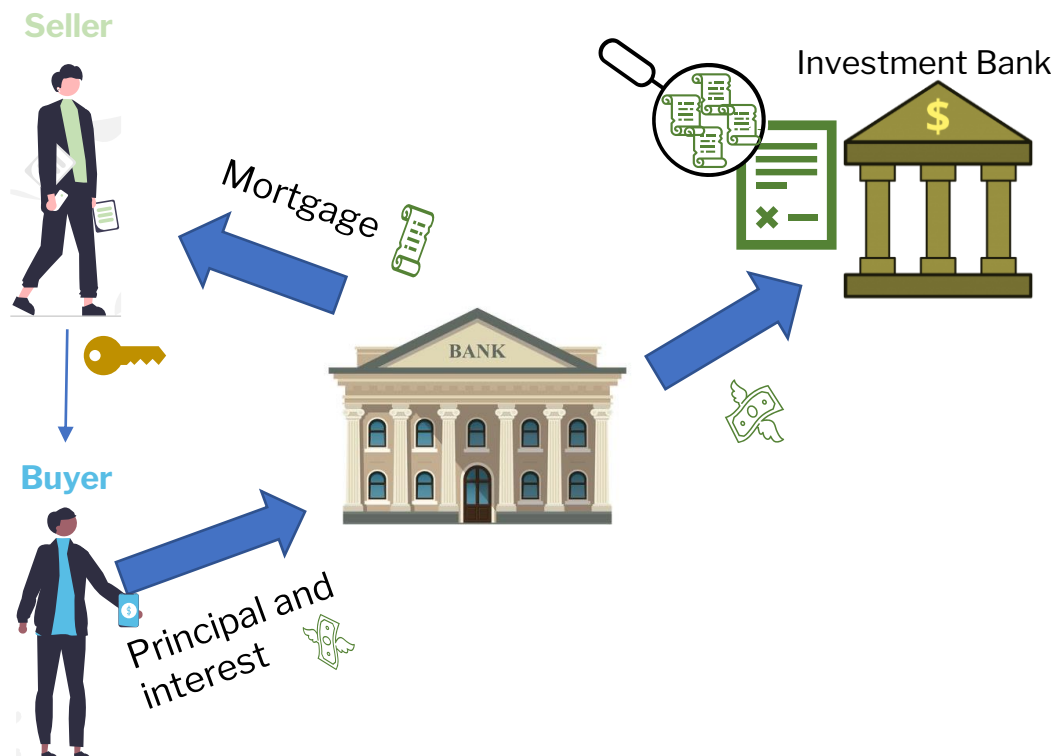
Mortgage-Backed Securities

Principle, Interest and Bonds covered in Lecture 6 on fixed income

What is a mortgage-backed security?

MBS – Mortgage-Backed Security

A type of bond representing an investment in a pool of real estate loans



Mortgage-Backed Securities

POSITIVE

Frees up bank's capital

POSITIVE

Bank makes money from originating + servicing mortgages and other fees

POSITIVE

Investors make money from interest and principal + change in value of MBS

Mortgage-Backed Securities

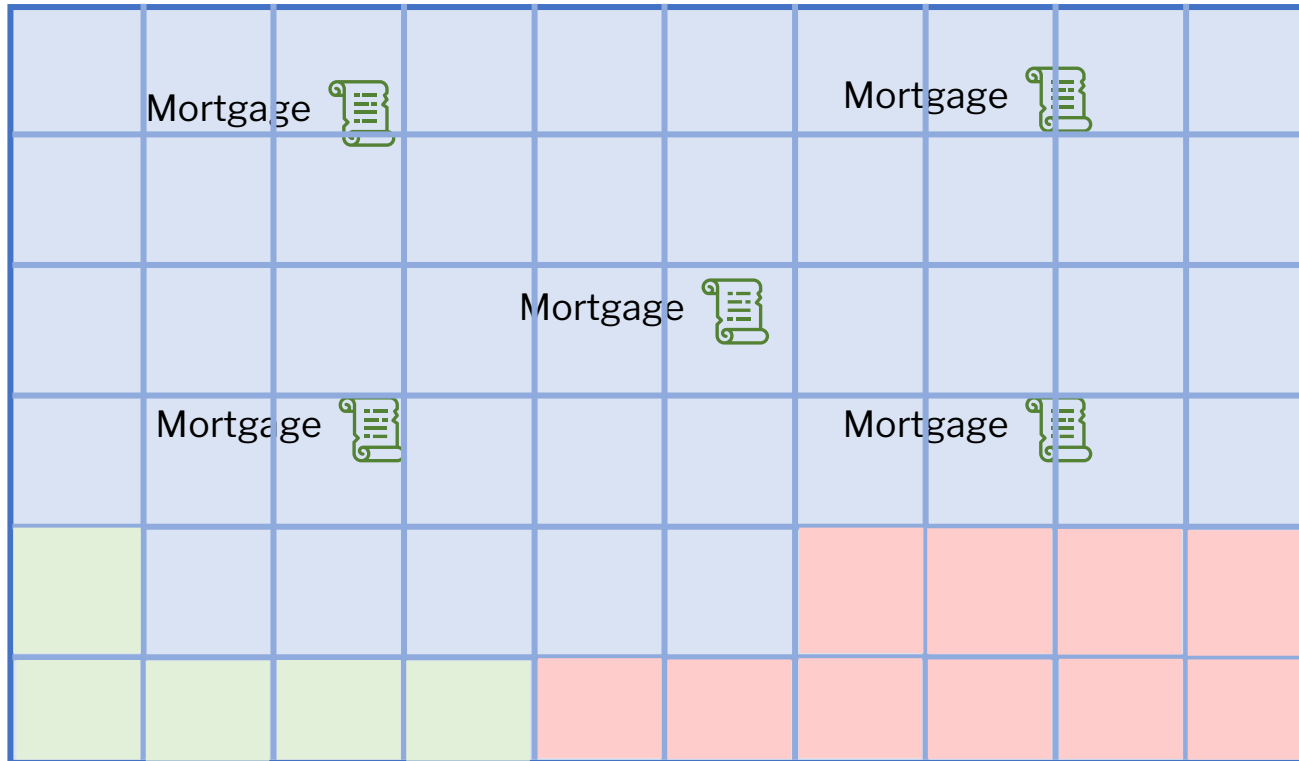
Principle, Interest and Bonds covered in Lecture 6 on fixed income

What is a mortgage-backed security?



MBS – Mortgage-Backed Security

A type of bond representing an investment in a pool of real estate loans



Someone may buy 5

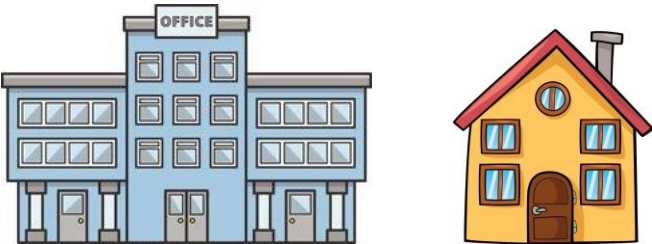
Someone may buy 10

iShares MBS ETF
Tracks results of an
index composed of
investment-grade MBS

What is a REIT?

👉 **REIT – Real Estate Investment Trust**
A company that owns, operates or finances income-producing real estate (RE)

👉 **PURPOSE:** Make the benefits of income producing real estate accessible to everyone.



Avg residential house price:
£299,249

Avg commercial real estate cost:
£2,190 / m²

Before REITs

Real estate sector *had* **high barriers** to entry

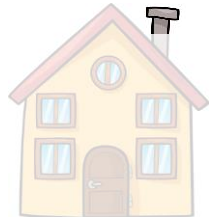
PROBLEM
Difficult to finance

PROBLEM
Missing out on benefit from RE investment

What is a REIT?

👉 **REIT – Real Estate Investment Trust**
A company that owns, operates or finances income-producing real estate (RE)

👉 Like “owning” a small part of the REIT’s assets



POSITIVE

Easy entrance to real estate market

POSITIVE

Low finance required

POSITIVE

Anyone can earn income through RE investment

Your notes

What qualifies as a REIT?

 **REIT – Real Estate Investment Trust**
A company that owns, operates or finances income-producing real estate (RE)

 To be classified as a REIT, a company must follow certain rules

Primarily own or finance real estate (75% AUM)

Have at least 100 share holders

Must distribute at least 90% of taxable income to shareholders as dividends

Be an entity taxable as a corporation

Majority of income must be earned from real estate (75% of income)

 If these rules are followed, a REIT pays **NO corporation tax** on profits

Your notes

What are the different types of REITs?

 **REIT – Real Estate Investment Trust**
A company that owns, operates or finances income-producing real estate (RE)

Equity REITs

Most common type

REIT itself **owns** income producing real estate

Revenue comes from rent

Mortgage REITs

REIT itself **finances** real estate owners

Can finance owners directly or indirectly (MBS)

Revenue comes from interest

Hybrid REITs

Combination of an **Equity** and **Mortgage** REIT

Owens income-producing real estate

Finance real estate owners

Your notes

REITs

What is one way of classifying REITs?

👉 You can classify REITs based on how the shares are traded.

Publicly Traded REITs

Shares listed on a national securities exchange

Regulated by the Securities and Exchange Commission (US)

Regulated by the Financial Conduct Authority (UK)

Publicly Non-Traded REITs

DO NOT trade on a national securities exchange

Private REITs

Not regulated by SEC (US)
or the FCA (UK)

Tend to be less liquid

Your notes

What are the advantages of a REIT?

Advantages of REITs

ADVANTAGE

Liquidity and ease of purchase

ADVANTAGE

No need for active management

ADVANTAGE

Relatively low entry price

ADVANTAGE

Diversification of portfolio

ADVANTAGE

Stable cashflow through dividends

ADVANTAGE

Access to commercial real estate

👉 Remember that these depend on the *types* of properties in question.

Your notes

What are the disadvantages of a REIT?

Disadvantages of REITs

DISADVANTAGE

Low growth

DISADVANTAGE

Works best as long-term investments

DISADVANTAGE

Yield is taxed as regular income

DISADVANTAGE

Management fees are included in price of REIT

DISADVANTAGE

Lack of control over returns of performance

👉 **AGAIN** Remember that these depend on the *types* of properties in question.

Your notes

Valuating REITs – Quantitative

Why are some traditional metrics a bad way of valuating REITs?

P/E = Share Price ÷ earnings per share

EPS = (Net income – dividends) ÷ outstanding shares

Net income = revenue – expenses

👉 **Expenses** must include depreciation of any assets the company holds.

👉 **BUT** properties tend to **appreciate** not depreciate.

	Year 1	Year 5	Year 10
Balance sheet, Assets:			
Real estate (gross assets)	1,000,000	1,000,000	1,000,000
(-) Minus accumulated depreciation	0	250,000	500,000
Net real estate (net assets)	1,000,000	1,000,000	1,000,000
Income statement			
Revenues	200,000	200,000	200,000
Expenses			
Operating Expenses	100,000	100,000	100,000
Interest Expense	40,000	40,000	40,000
Annual depreciation	50,000	50,000	50,000
Total expenses	190,000	190,000	190,000
Net income (revenue – expenses)	10,000	10,000	10,000
Funds from operation (FFO) (FFO = net income + depreciation + etc)	60,000	60,000	60,000

Valuating REITs - Quantitative

What is a metric that can be used instead?

$$\text{FFO} = \text{Net Income} + \text{Depreciation} + \text{Amortization} - \text{Gains on Sales of Property} - \text{Interest Income} + \text{Losses on Property sales}$$

👉 **FFO – Funds From Operations**

👉 Cash flow from the REIT's operations.

👉 Adds depreciation back to net income.

Your notes

Balance sheet, Assets:

	Year 1	Year 5	Year 10
Real estate (gross assets)	1,000,000	1,000,000	1,000,000
(-) Minus accumulated depreciation	0	250,000	500,000
Net real estate (net assets)	1,000,000	1,000,000	1,000,000

Income statement

Revenues	200,000	200,000	200,000
Expenses			
Operating Expenses	100,000	100,000	100,000
Interest Expense	40,000	40,000	40,000
Annual depreciation	50,000	50,000	50,000
Total expenses	190,000	190,000	190,000
Net income (revenue – expenses)	10,000	10,000	10,000
Funds from operation (FFO) (FFO = net income + depreciation + etc)	60,000	60,000	60,000

Valuating REITs - Quantitative

What does FFO show?

$$\text{FFO} = \begin{array}{c} \text{Net Income} \\ + \\ \text{Depreciation} \\ + \\ \text{Amortization} \\ + \\ \text{Losses on Property sales} \end{array} - \begin{array}{c} \text{Gains on} \\ \text{Sales of Property} \end{array} - \text{Interest Income}$$

👉 **FFO – Funds From Operations**

👉 Gives an idea of cash generated by a REIT

👉 Price/FFO is like the P/E ratio but for REITs

👉 Can be used to compare different REITs

Valuating REITs - Quantitative

What does AFFO show?

👉 FFO misses part of capital expenditure required to acquire or maintain property
e.g. painting, plumbing, associated fees etc.

$$\text{AFFO} = \text{FFO} + \text{Rent increases} - \text{Capital expenditures} - \text{Routine maintenance amounts}$$

👉 **AFFO** stands for **A**ddjusted **F**unds **F**rom **O**peration

Gives **full** picture of company
(residual cash flow for shareholders)

Considers additional costs and
income sources

Overview of future capability to
pay dividends

👉 AFFO may **NOT** be useful in *all* circumstances!

Your notes

Valuating REITs

What does NAV show?

NAV – Net Asset Value

- ☞ A method of modelling REIT's future performance
- ☞ Assess fair market value of REIT's Real Estate assets
- ☞ Factors in future rent and property appreciation
- ☞ Favoured method of REIT valuation as it relies on real estate market prices to determine value
- ☞ Can be used to compare to similar size REITs

Valuating REITs - Qualitative

What are some qualitative methods for valuating REITs?

Types of Properties

☞ **Diversification** in property sectors e.g.

Residential

Retail

Office.

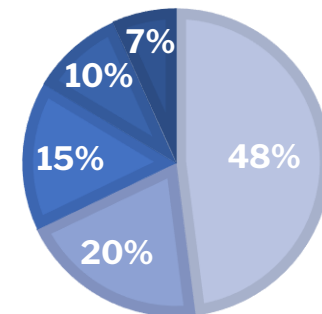
☞ Used to predict performance in economic climates

☞ Examples include recession, property boom etc.

Your notes

AEWU REIT SECTOR WEIGHTINGS

■ Industrial ■ Offices
■ Retail Warehouse ■ Standard Retail
■ Leisure



Valuating REITs - Qualitative

What are some qualitative methods for valuating REITs?

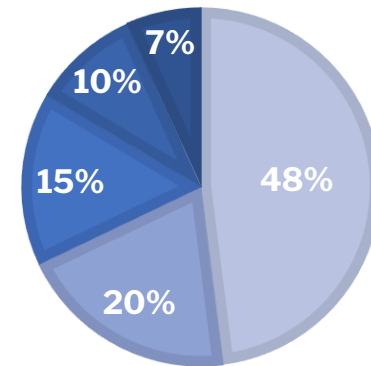
Types of Properties

☞ **Diversification** in property sectors
(residential, retail, office etc)

☞ Way of predicting performance in certain
economic climates

AEWU REIT SECTOR WEIGHTINGS

■ Industrial ■ Offices ■ Retail Warehouse
■ Standard Retail ■ Leisure



Your notes

Valuating REITs - Qualitative

What are some qualitative methods for valuating REITs?

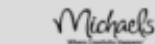
Types of Tenants

- ☞ Percentage of portfolio made up of big companies (e.g. Walmart)
- ☞ Large pool of tenants so REIT is not adversely affected if one tenant were to perform negatively
- ☞ Reliability of tenants
- ☞ This works on a case-by-case basis.

Your notes

Valuating REITs - Qualitative

What are some qualitative methods for valuating REITs?

Retailer Category	% of Rent	Key Brands (ii)
Grocery / Pharmacy / Liquor	19.9%	     
Essential Personal Services	14.3%	     
Specialty Retailers	11.9%	     
Value Retailers	10.6%	   
Furniture & Home	8.9%	     
Other Personal Services	8.0%	  
Quick Service Restaurants	7.1%	     

NOTE: A diversity of business types exist within REITs.

Your notes

Valuating REITs - Qualitative

What are some qualitative methods for valuating REITs?

Geographic location

👉 **Geographically diverse.**
Different properties in multiple areas

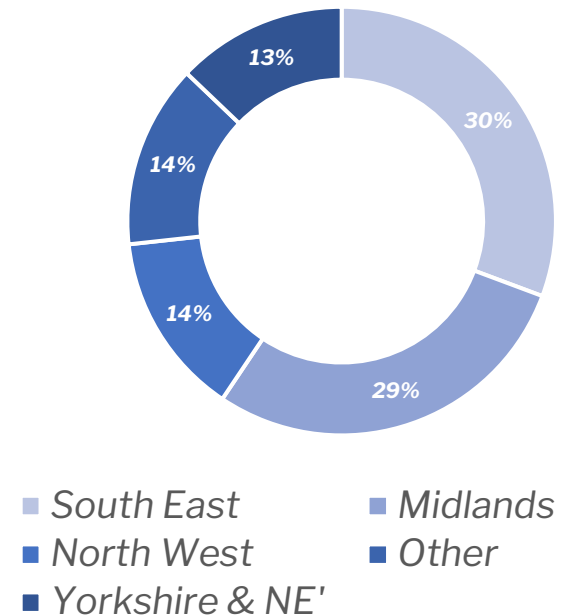
👉 Spread within a country & international reach.

Occupancy and Rent

👉 **Occupancy Rate** – proportion of properties occupied by tenant's vs property owned

👉 **Average lease length**
A measure that can reflect stability of REIT

Urban logistics
Geographic locations



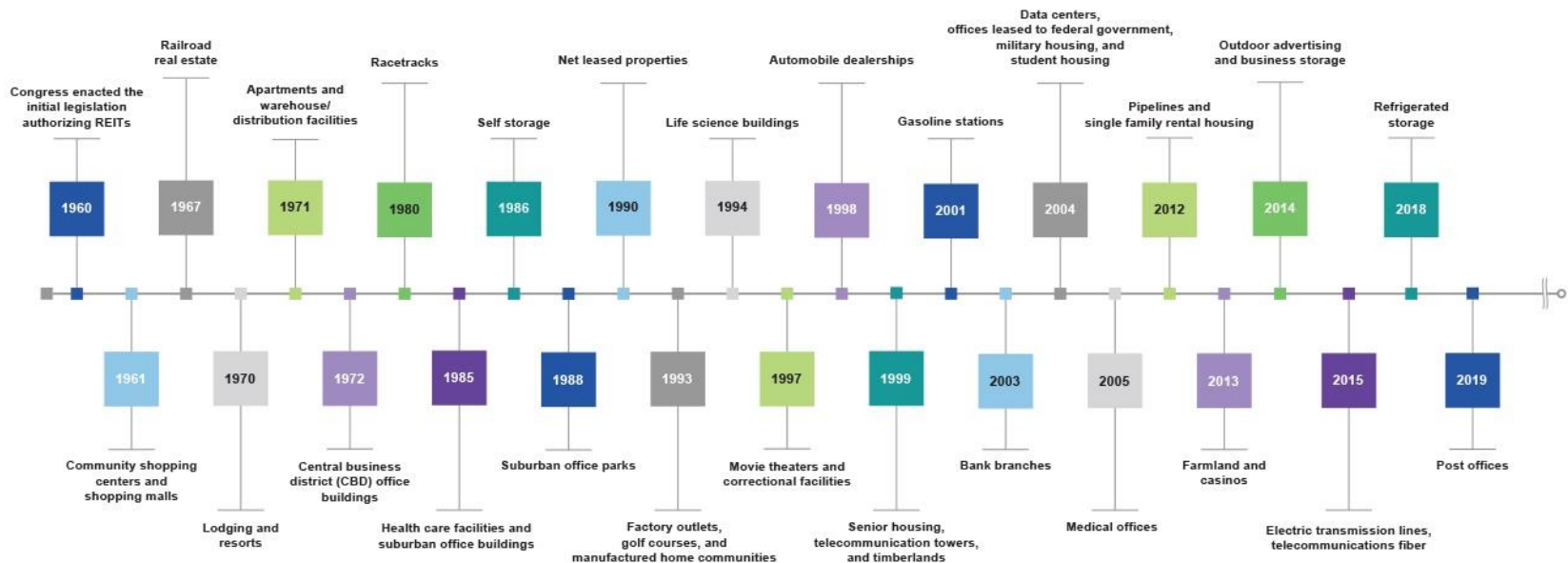
Application of REITs

REITs were established by Congress in **1960** so all investors can access income-producing real estate.

Intention: For all people to invest in high quality real estate

Today, approximately **145 million Americans** (~44% of American households) own REIT stocks/
Directly or indirectly through mutual funds, ETFs, or target date funds

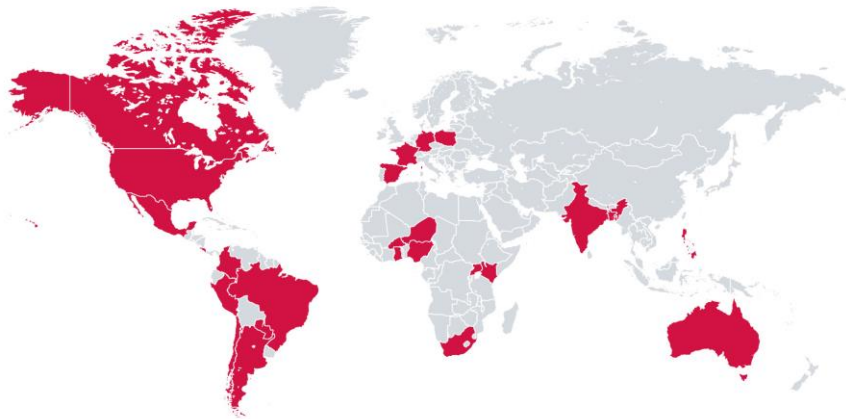
Timeline of REIT Listings by Property Type



Above shows a timeline for the properties that were included in REIT's

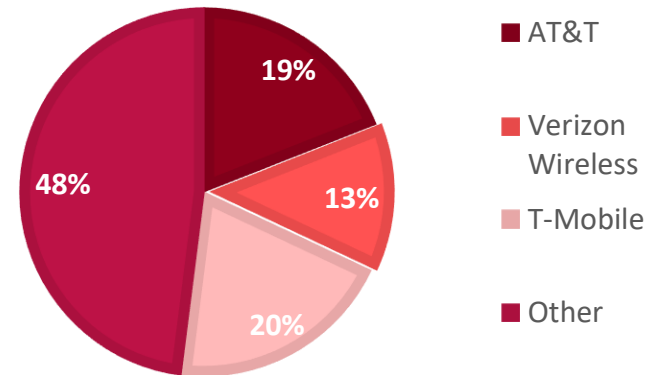
American Tower Corporation (NYSE: AMT)

- American Tower Corporation is the world's **largest REIT** with a market cap of **\$117.97 Billion**.
- It is a leading independent owner, operator and developer of wireless and broadcast **communications** real estate.
- Global portfolio of approximately **222,000** communication sites in **21 countries**



American Tower Corporation's global reach

AMT REVENUE SHARE BY TENANT WORLDWIDE



Diversification of tenants reduced risk for AMT

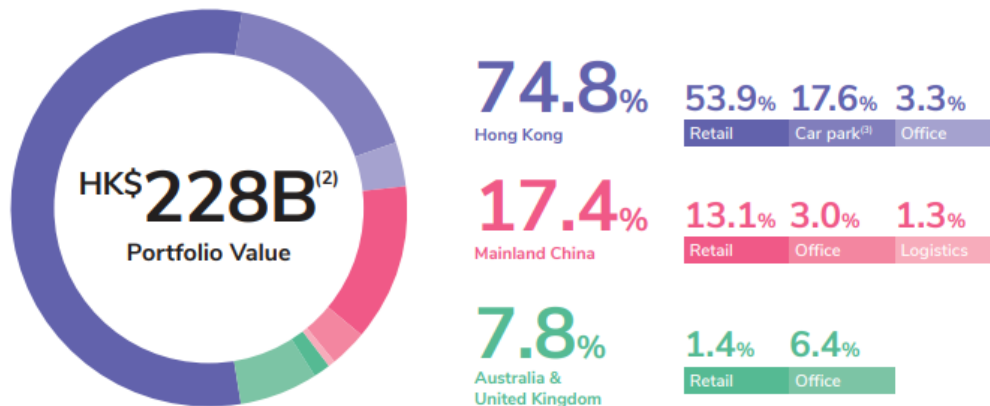


AMERICAN TOWER®

Link REIT (HKSE: 0823.HK)

- Largest **hybrid REIT** in Asia
- Invests and manages a diversified portfolio comprising of retail facilities, car parks, offices and logistics centers.
- Mainly based in Hong Kong with around 129 investments locally and only 22 investments outside Hong Kong

Portfolio Value



Notes:

Link REIT portfolio breakdown:

https://www.linkreit.com/-/media/corporate-website/investor-relations/financial-report/2022/annual-report-2122_e_book-1.pdf



Annaly Capital Management (NYSE: NLY)

- World's largest mortgage REIT with \$89 billion in total assets
- \$13 billion of which are permanent capital – investments for an undefined period of time
- Primary focus on borrowing via short term security repurchase agreements and reinvesting the proceeds in asset backed securities

Divided yield of around 10.68%

- For every £100 invested, shareholder will receive £10.68 in dividends each year

Paid out \$1.4 billion in **dividends** to shareholders

Delivered total returns of 925% since IPO outperforming the S&P 500



Annaly Capital Management total returns since IPO vs S&P 500:

<https://www.annaly.com/~media/Files/A/Annaly-V3/documents/Earnings%20Docs/2022/annual-report-2021.pdf>

ANNALY®

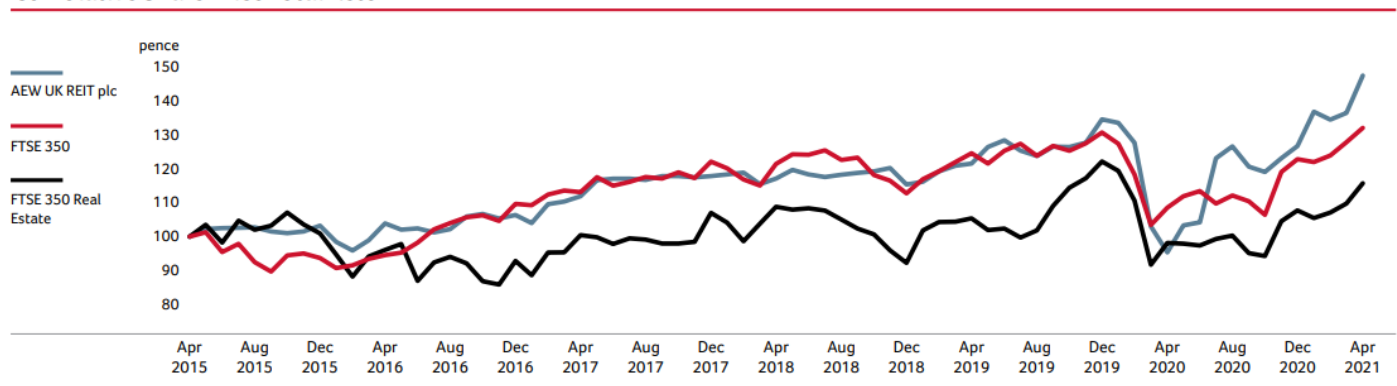
AEW UK REIT (LSE: AEWU)

- Largest REIT in the UK with a **NAV of £ 200.4 million**
- Invests predominantly in smaller **commercial** properties (< £15 million) on shorter occupancy leases
- Portfolio locations across the UK
- Tenants include Poundland Ltd, Odeon Cinemas and Wilko Retail Ltd.

Divided yield of around 7.17%

- For every £ 100 invested, shareholder will receive £ 7.17 in dividends each year
- For comparison average dividend yield of a FTSE 250 is 4%

Cumulative Share Price Total Return



Graph from Annaly Capital Management Annual report

https://www.annualreports.com/HostedData/AnnualReportArchive/a/LSE_AEWU_2021.pdf



Let's resume...



Joseph Guillon

Lecturer

*S&T (Summer 2023), Morgan Stanley
Fast Track, Amplify Trading (2022)
Secretary, Imperial Blockchain Group (2022-23)
Analyst, Transport @ QT Capital (2021 – 22)*

Lecture 7

Commodities

Introduction to Commodities

What do we actually mean by **commodities**?

☞ Simple: basic goods possessing **fungibility**.

☞ Raw materials used for human standard of living → *basic good*.

Examples

Basic Goods

Natural Gas

Rice

Steel

☞ A basic good that can be interchanged without loss of quality → **fungible**.

Peruvian Gold is identical to Australian Gold (for the same purity) → *fungible* → same prices.

Your notes

Classifying Commodities

How do we class commodities given their properties?

👉 Commodities are either *hard* **OR** *soft*.

Hard commodities

Extracted from the Earth

Energy

Metals

*Metals can be
further divided...*

Precious
Metals

Base
Metals

Soft commodities

Grown naturally

Agricultural
Commodities

Livestock

Meat

This depends on how the metal derives its value.

Your notes

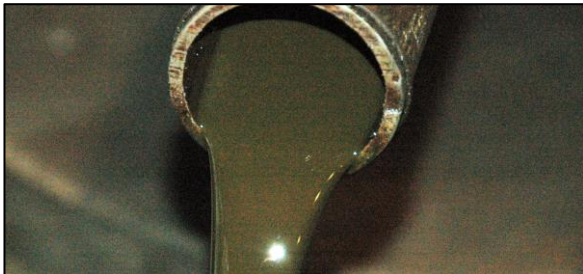
Energy Commodities

Let's discuss the first type of commodity... **Energy**.

☞ Energy commodities can be used to release energy.

Energy commodities

Crude Oil



James St. John // Flickr

Natural Gas



Kwon Junho // Unsplash

Coal



Joey Harris // Unsplash

Your notes

Precious Metals

Another type that you'll be familiar with is **Precious Metals**.

👉 Precious metals are *rare* and have *specific* uses.

Precious Metals

Gold



James St. John // Flickr

Silver



Jurii // Images of Elements

Platinum



James St. John // Flickr

Palladium



Jurii // Images of Elements

Your notes

Base Metals

What about **base metals**?

👉 Base metals are *common* in nature but are used widely.

Base Metals

Steel



Jatinsanghvi // Wikimedia

Nickel



Images of Elements // Wikimedia

Aluminium



Jurii // Images of Elements

Copper



Jonathan Zander // Wikimedia

Your notes

Agricultural Commodities

So, what is there to know about **agricultural commodities**?

👉 Agricultural commodities are harvested for food **OR** fuel.

Agricultural commodities

Corn

Rice

Wheat

Coffee

Sugar

Cocoa



Andrew Butkoi // Wikimedia



Anna Frodesiak // Wikimedia



Simon Waldherr // Wikimedia



Freestock.ca // Wikimedia



422737 // Pixabay



dfataustralianaid // Flickr

Your notes

Livestock and Meat Commodities

But what about commodities that are alive? **Livestock** and **Meat Commodities**!

👉 Livestock and Meat Commodities are animals raised for labour **OR** food.

Livestock and Meat Commodities

Live Cattle



John Comloquoy / Beef cattle on Eefie Hill / CC BY-SA 2.0

Feeder Cattle



Kevin McManus // Flickr

Juicy Bacon

Also known to traders as
'Frozen Pork Bellies'

CME Index Ticker
GPB

Your notes

Commodity Market Players

Who gets involved in commodity markets?

☞ Two types of players are involved:

Hedgers

Commercial players

Producer **OR** Consumer
of the commodity

Aim to reduce price *volatility*
of the commodity they trade

Speculators

Standard traders **OR** investors

NEITHER Producer **OR** Consumer
of the commodity

Aim to make profit
from commodity they trade

Your notes

Commodity Market Players

How do market players ensure they achieve their goals?

👉 They use instruments and/or strategies you are familiar with in previous SEC Lecturers...

Hedgers

Buy future contracts
'Lock-in' Prices

Buying airplane fuel contracts *in advance*
Airline (Buy-Side) Hedging

Sell a futures contract *at an agreed price*
Farmer (Sell-Side) Hedging

Speculators

Observe market **fluctuations**

Savvy Retail Investors (e.g. You)

A commodities trader
e.g. at Morgan Stanley

Your notes

*TLDR: Hedgers **try** to protect themselves from price fluctuations **BUT** speculators try to profit off it.*

Why invest in commodities?

If commodities' prices are very volatile, why invest in them?

👉 Commodities looks set to grow in the coming years...

Population Growth

15th Nov, 2022: 8 Billion
2055: 10 Billion

More people mean more demand for resources.

Urbanisation

68% of people will live in cities by 2050 (UN).

Urban lifestyles consume **more** resources than rural lifestyles.

Industrialisation

The BRICS play a majority of world economic growth.

They increase commodity demand in industrialisation.

Prices of commodities will **rise**.
via Supply and Demand theory (Lecture 2)

Your notes

Why invest in commodities?

There are more reasons that you could invest in commodities as an investor...

Hedge against inflation

Commodities benefit from inflation.
Ironically, increases in commodity prices drive inflation themselves.

Safe haven of value

Precious metals e.g. gold or silver are financially safe.
Their prices may increase e.g. after COVID-19, Ukraine War, Currency falls etc.

Inelastic Nature

Inelastic goods' demand **don't** react to price changes.
Even if their prices increase, people still buy them e.g. food, fuel, gas etc.

Portfolio Diversification

Commodities *used* to hedge against market downturns.
Equity and commodities markets are **sometimes** correlated (see notes).

 Commodities markets can be explained by macroeconomic theory (Lecture 2).

Your notes

Commodities' Focus: Oil

*Let's focus on a specific type of commodity... **Oil**.*

👉 Before an investment is made into a commodity, fundamental questions need to be asked:

Where are the **reserves** located?

What **companies** are exploiting it?

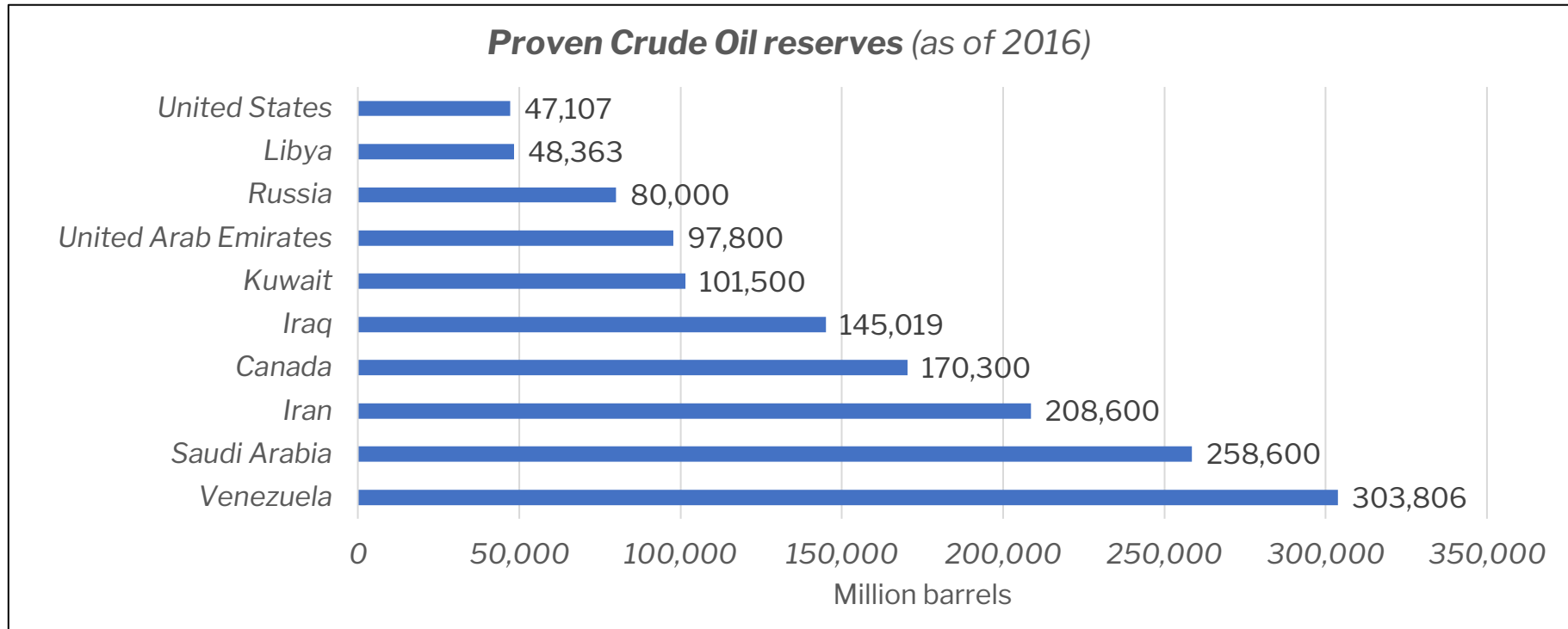
What are its **uses**?

What has been the historic performance?
What is the outlook for the future considering current supply and demand trends?

How is it affected by macroeconomics?

Commodities' Focus: Oil

Where are the crude oil **reserves**?

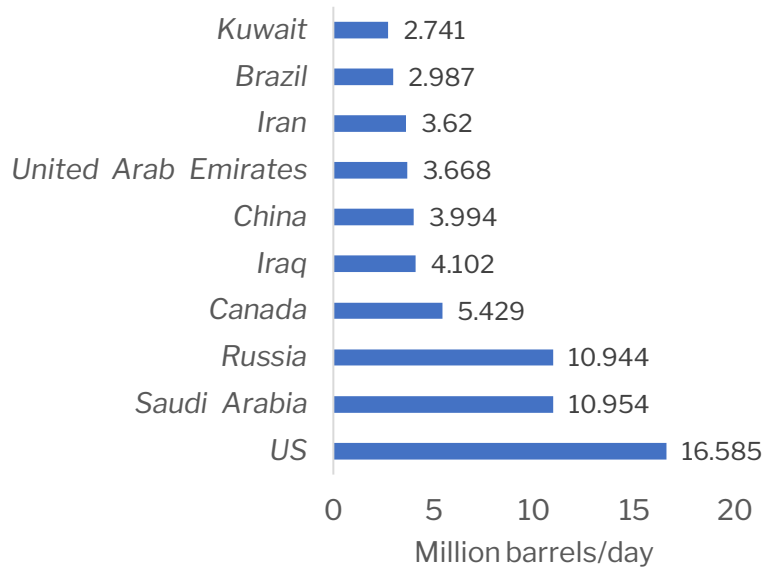


Top 10 Countries by Proven Oil Reserves (US Energy Information Administration, 2016)

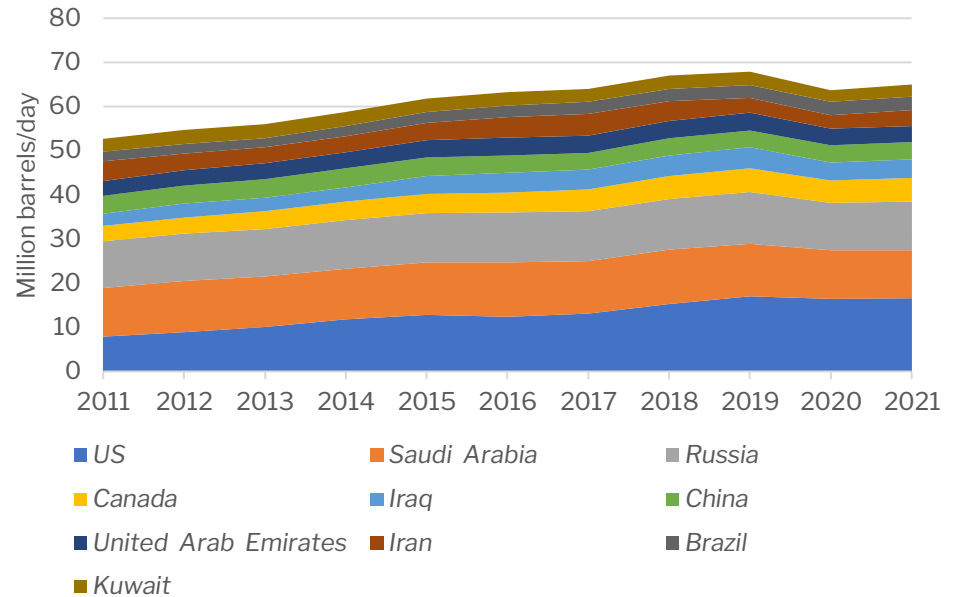
Commodities' Focus: Oil

Where is crude oil *produced*?

**Top 10 Countries
by daily Crude Oil production (in 2021)**



**Top 10 Countries
by Crude Oil production (2011-2021)**

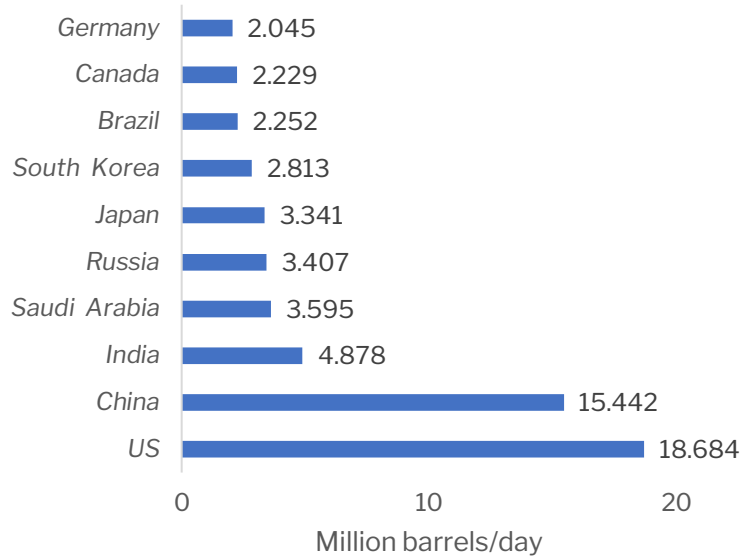


👉 Notice the total production increasing over time (except 2020...)

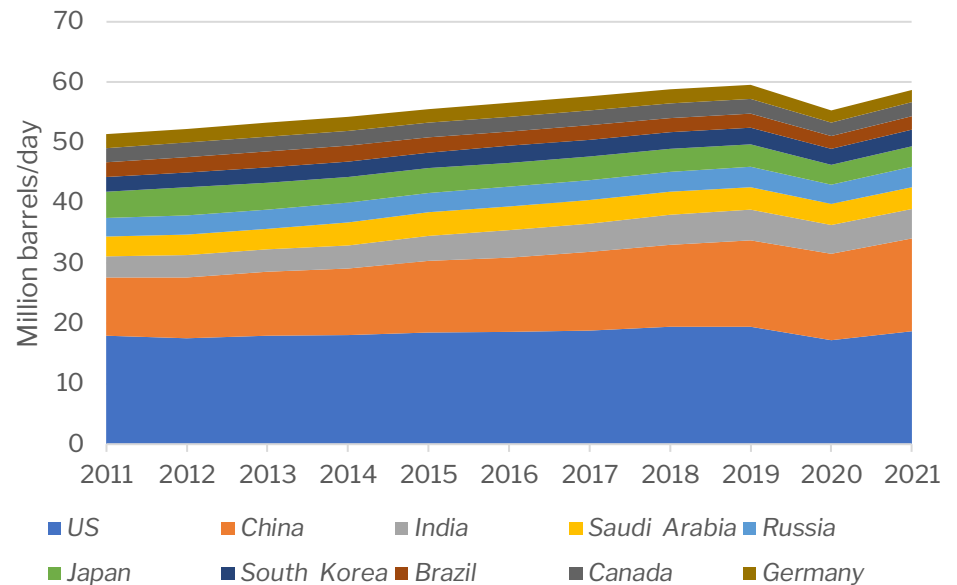
Commodities' Focus: Oil

Where is crude oil **consumed**?

**Top 10 Countries
by Daily Consumption (in 2021)**



**Top 10 Countries by Daily Consumption
(2011-2021)**

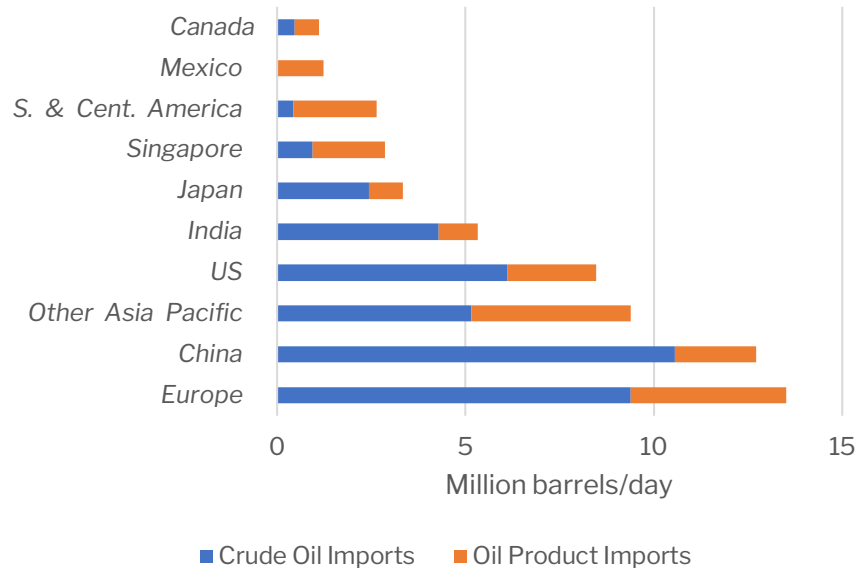


👉 Notice the total consumption increasing over time (except 2020...)

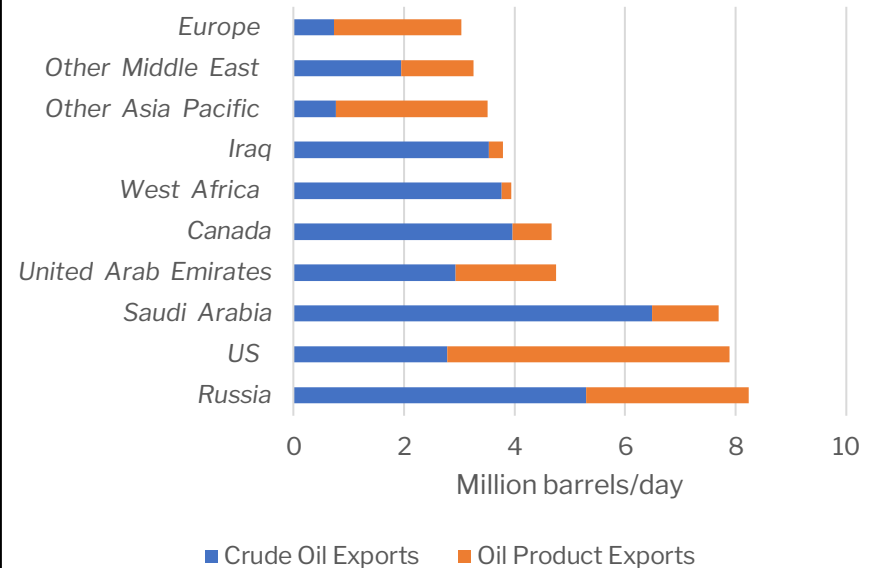
Commodities' Focus: Oil

Who are the crude oil **importers** and **exporters**?

Top 10 importers of Crude Oil and its derivatives (in 2021)



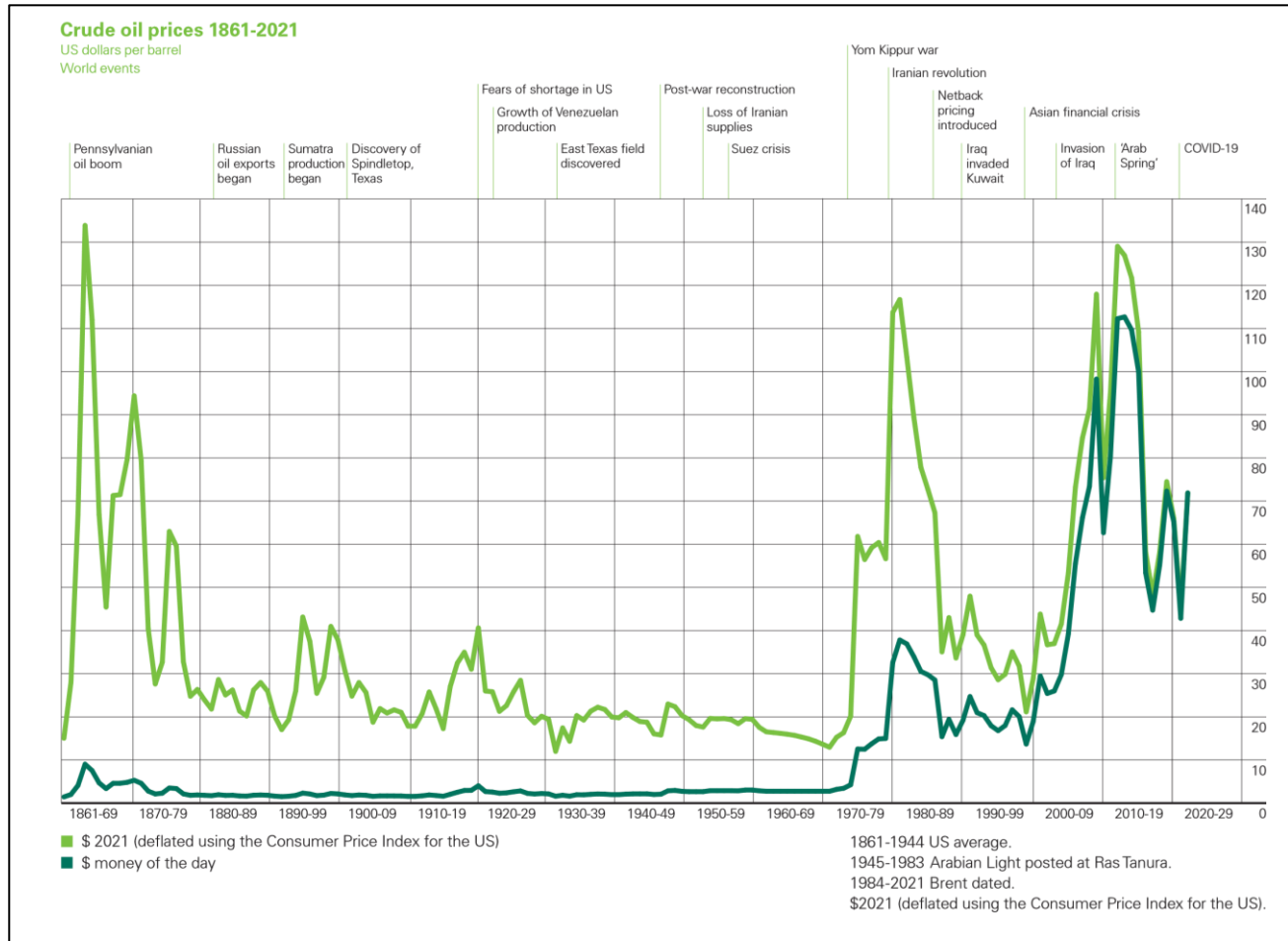
Top 10 exporters of Crude Oil and its derivatives (in 2021)



👉 Notice the difference in list between oil *importers* and *exporters*!

Commodities' Focus: Oil

What are the **historic prices** of crude oil?



A History of Crude Oil Prices 1861-2021 (BP Statistical Review, 2022)

Methods to invest in commodities

Now we've discussed what they are, **how** would someone invest in commodities?

☞ There are several ways for investors to invest in commodities...

Derivatives

Futures

Forwards

Options

Indexes

ETFs

Mutual Funds

Equity

Stocks

MLPs

Specialists

CTAs

CPOs

☞ All these methods are discussed in the following slides.

Your notes

Derivatives

How do derivatives play a part in trading commodities?

☞ Futures allow you to gain exposure to commodities whilst reducing risk.

Commodity
Exchanges

CME

Chicago Mercantile Exchange

LME

London Metal Exchange

☞ Forwards are used more by *hedgers* as they are OTC → *more flexible & tailor-made*

☞ Options can also be used (see Lecture 5).

☞ The derivative choice depends on the **intention** of the contract buyer.

Your notes

Indexes

You saw indexes in Lecture 3... how can they help to trade commodities?

☞ Indexes help measure the overall 'health' of the commodities market.

☞ **N.B.** Commodity indexes measure a *basket* of underlying commodities' futures contracts.

☞ Futures expire after a time so they are 'rolled' i.e. close the current future and buy a new one.

**S&P Goldman Sachs
Commodity Index**

Reuters/Jefferies **Commodity
Research Bureau Index**

The **Dow Jones** – **AI
Commodity Index**

S&P GSCI

CRB

DJCI

☞ You cannot invest into a commodity index directly.
You can invest in a fund that tracks the index performance.

You've seen **most** of this before in Lecture 3 with Adam (Equities)

ETFs and Mutual Funds

How do ETFs and Mutual Funds come to the rescue?

👉 ETFs and Mutual Funds both track the performance of an index.

👉 ETFs track an index's performance but trade like a stock.

👉 Mutual Funds pool investors' capital to buy securities.
They are actively managed by a fund manager.

👉 Investors can also invest in **index futures**.
A futures contract that tracks the performance of an index.

👉 Index futures are more expensive → more '*capital intensive*'.

Your notes

*You've seen **most** of this before in Lecture 3 with Adam (Equities)*

Stocks and MLPs

Nearly there to the end of the SEC... what are stocks and MLPs?

👉 Buying shares of commodity *companies* gives indirect exposure to the commodities market.

For exposure to the oil markets, buy an oil producing company e.g. BP, Shell etc.

👉 **Master Limited Partnerships (MLPs)** do not pay corporate income taxes.
Commodities' companies are structured as MLPs.

👉 MLPs can be traded on public exchange through shares.

👉 MLPs have a pass-through tax structure.
Investors receive untaxed earnings so MLP earnings are only taxed **ONCE**.

Your notes

CTAs and CPOs

How do specialists help gain exposure to commodities?

👉 **Commodity Trading Advisors (CTAs)** specialise in trading in *futures* markets.

👉 **Commodity Pool Operators (CPOs)** manages a fund *pool* to trade into commodity futures..

👉 Investing into CPOs has advantages:

A larger pool of funds → more futures' contracts purchased → **Diversification**

More funds → more **leverage** → greater returns

CPOs are Limited Liability Partnerships → You only lose the money you put in → **Limited Risk**

*Due diligence with CTAs and CPOs is **strongly advised** e.g. track record, disciplinary actions, fees, minimum sum etc.*

Commodities Market Indicators

The end! What indicators are used to track the state of the commodities market?

👉 Looking at prices of commodity futures is **insufficient** when deciding to invest in commodities.

👉 Several crucial market indicators should be used:

CPI

Consumer Price Index

GDP

Gross Domestic Product

EIA

Inventory Reports

Federal Funds Rate

London Gold Fix

Nonfarm Payrolls

PMI

Purchasing Managers Index

Reuters/Jefferies
CRB Index

U.S. Dollar

WTI

Crude Oil

Your notes